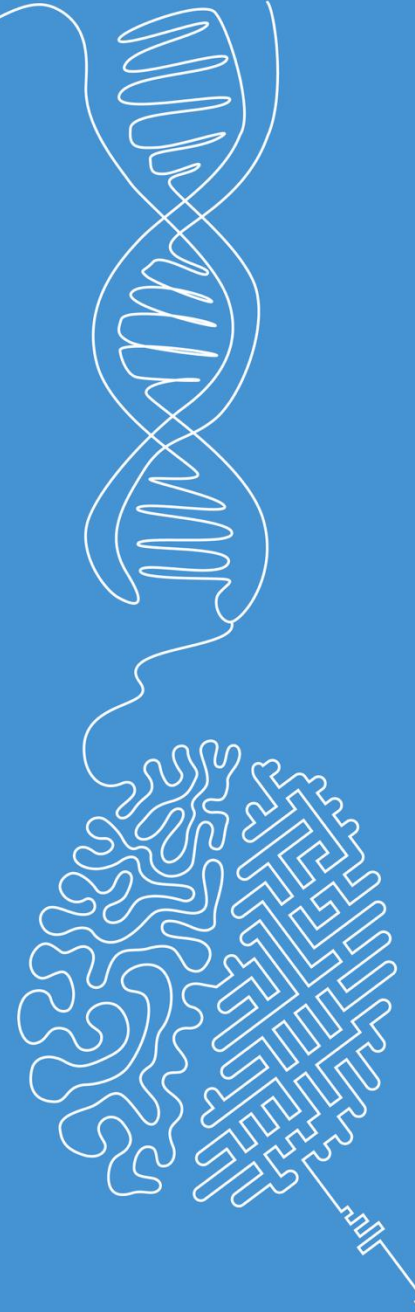


Fourier Transform in 2D

Image and Signal Processing

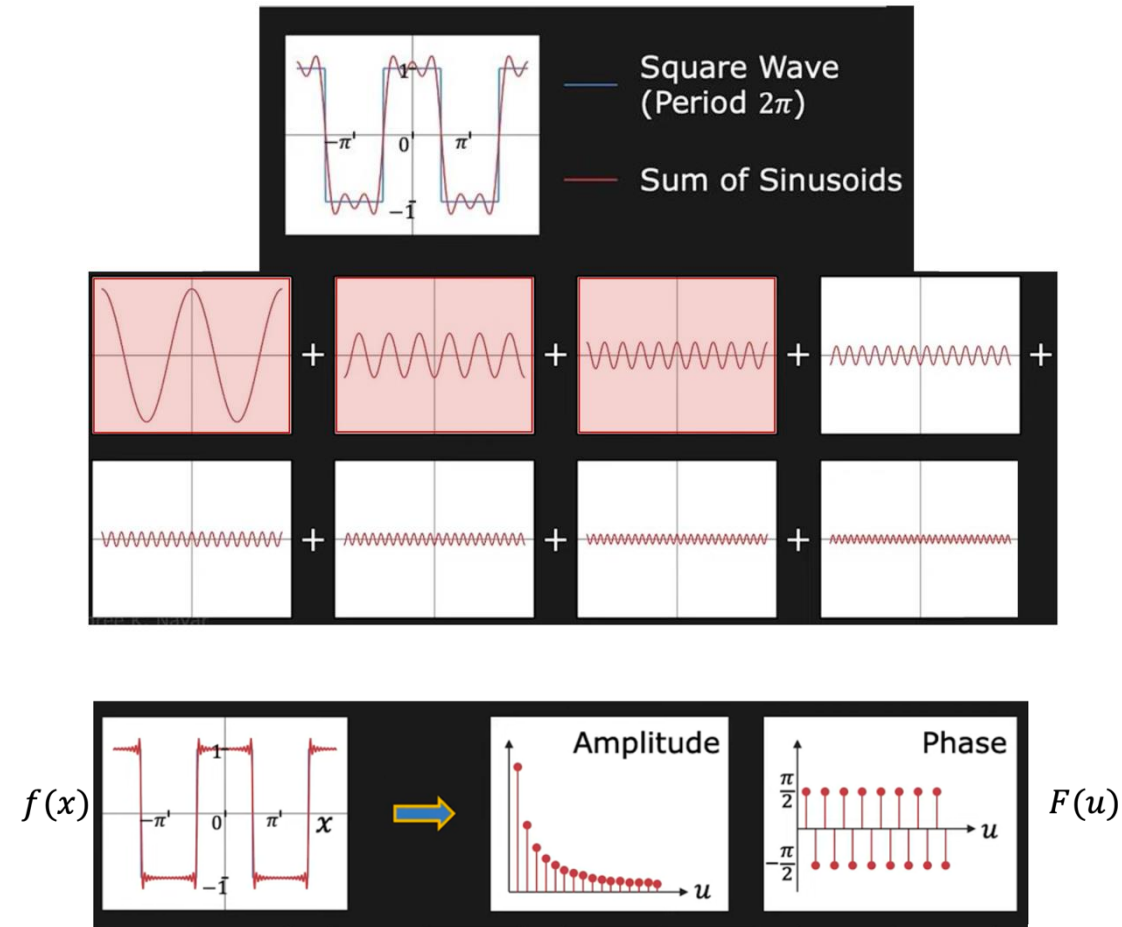
Norman Juchler



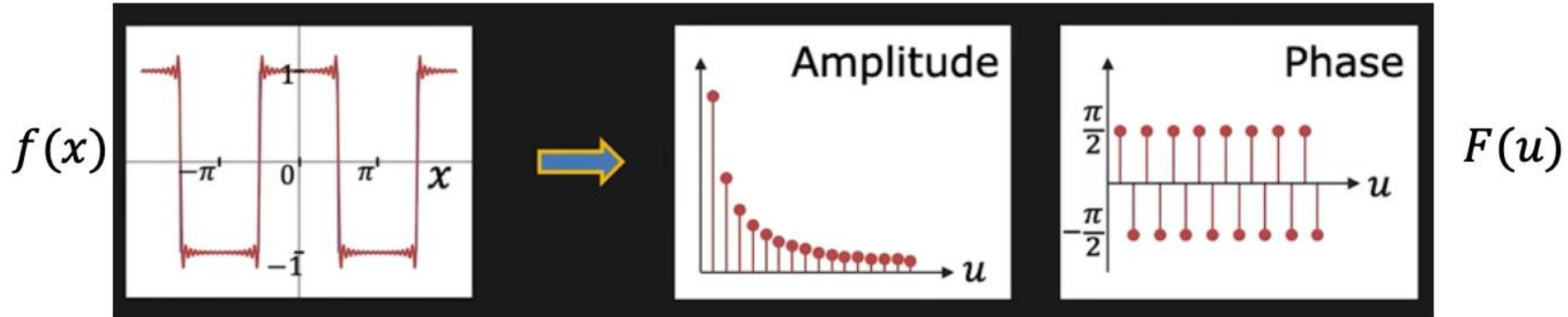
Recap: Fourier series / Fourier transforms

What we have already seen:

- Any periodic signal can be represented as a weighted sum of sine (and cosine) functions.
- This idea generalizes to non-periodic signals via the Fourier Transform
- The Fourier Transform generally produces complex-valued results $\rightarrow$ typically visualized using magnitude (amplitude) and phase spectra
- In practice, we compute the discrete Fourier transform (DFT) using the fast Fourier transform (FFT)



Fourier series / Fourier transforms

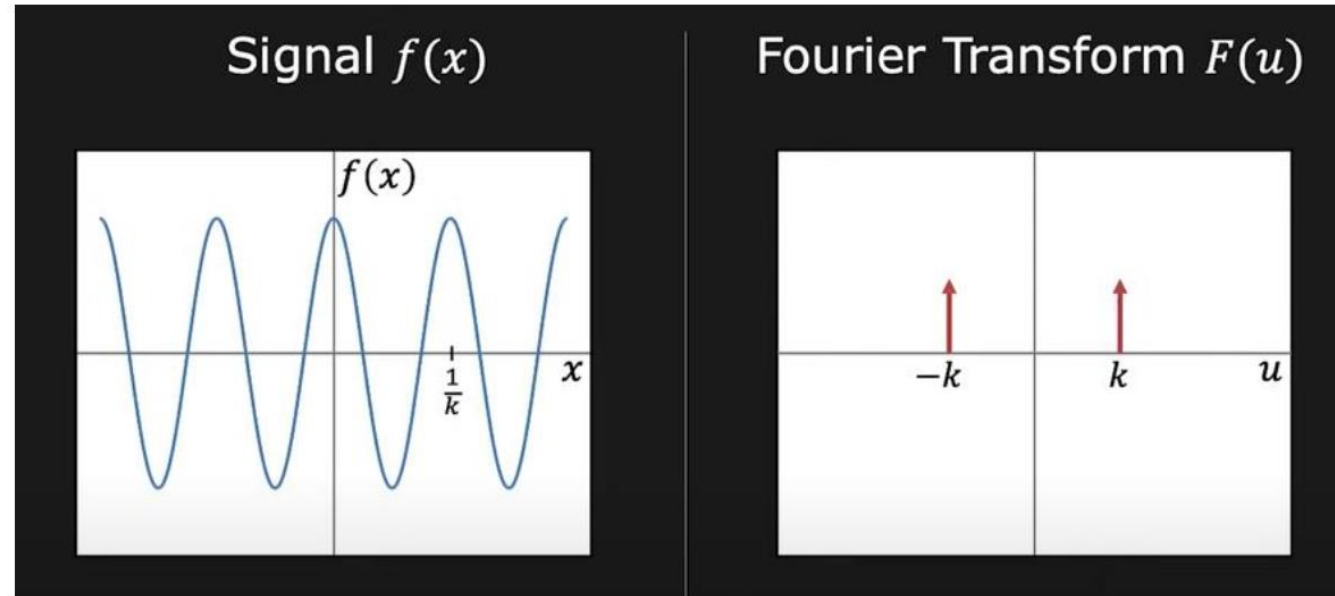


$$f(x) \longrightarrow \boxed{\text{FT}} \longrightarrow F(u)$$

$$f(x) \longleftarrow \boxed{\text{IFT}} \longleftarrow F(u)$$

Fourier series / Fourier transforms

Example

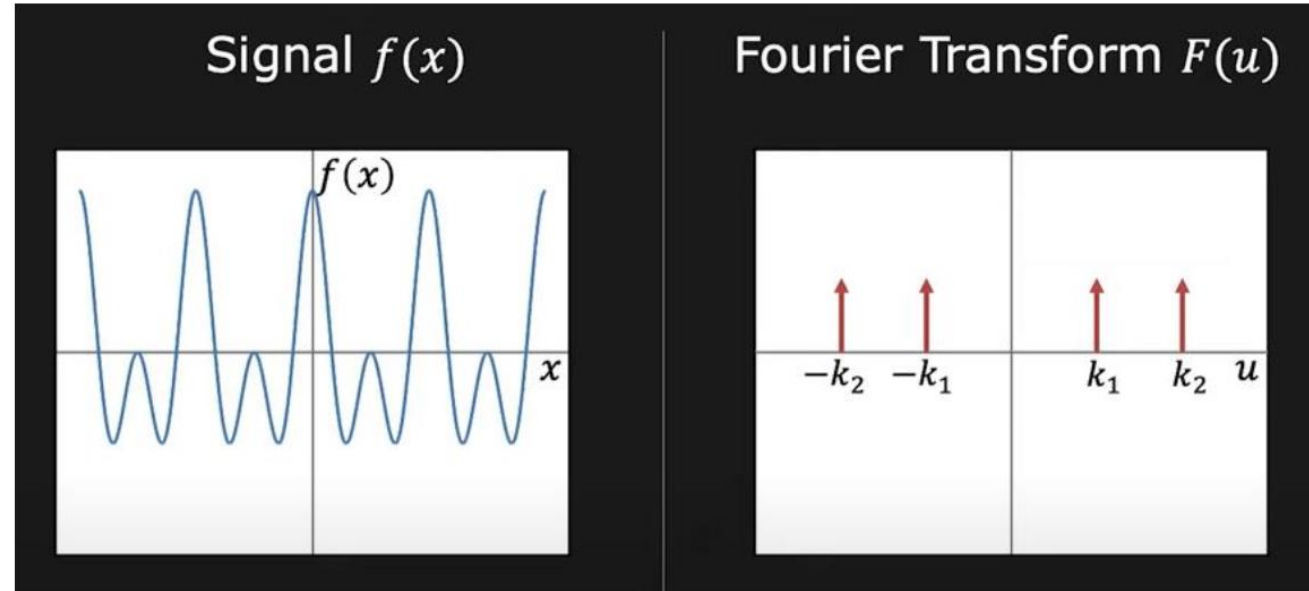


$$f(x) = \cos(2\pi kx)$$

$$F(u) = \frac{1}{2} [\delta(u + k) + \delta(u - k)]$$

Fourier series / Fourier transforms

Example



$$f(x) = \cos(2\pi k_1 x) + \cos(2\pi k_2 x)$$

$$F(u) = \frac{1}{2} [\delta(u + k_1) + \delta(u - k_1) + \delta(u + k_2) + \delta(u - k_2)]$$

Generalization to 2D

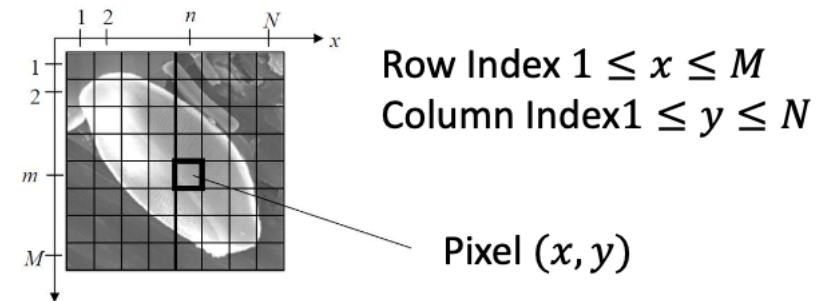
- The Fourier transform – and its discrete variants – can be naturally extended to two dimensions.

Discrete Fourier Transform (DFT)

$$F[u, v] = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f[x, y] e^{-\frac{i2\pi(ux)}{M}} e^{-\frac{i2\pi(vy)}{N}}$$

Inverse Discrete Fourier Transform (IDFT)

$$f[x, y] = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F[u, v] e^{\frac{i2\pi(ux)}{M}} e^{\frac{i2\pi(vy)}{N}}$$



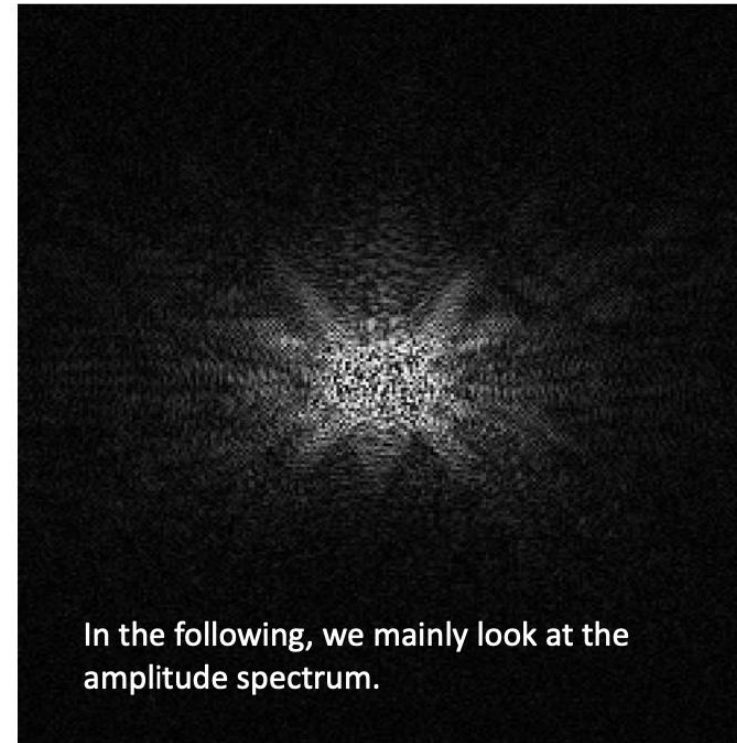
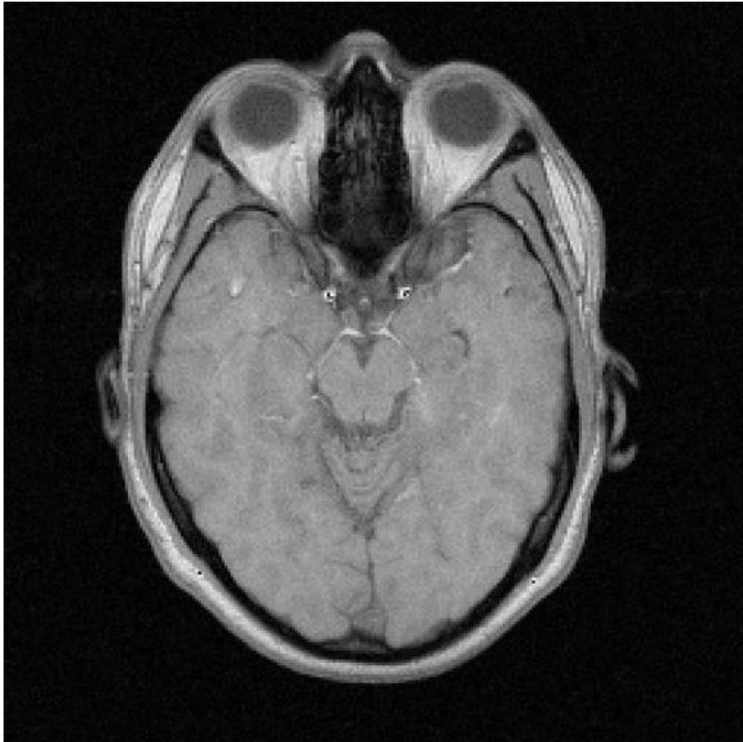
x, y : Image coordinates

u, v : Frequencies

$$i = \sqrt{-1}$$

This is called a **linear, bijective** transformation → **no image information is lost.**

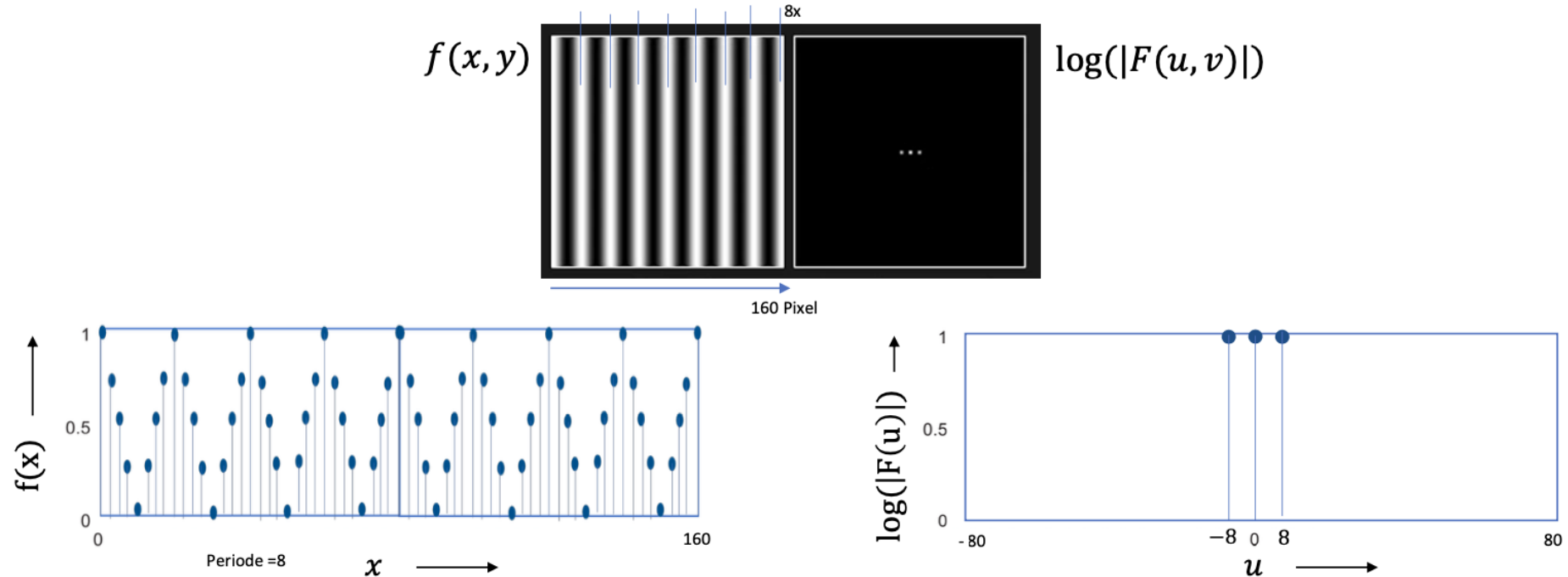
Frequency domain representation of an image



Frequency domain representation of an image

Examples:

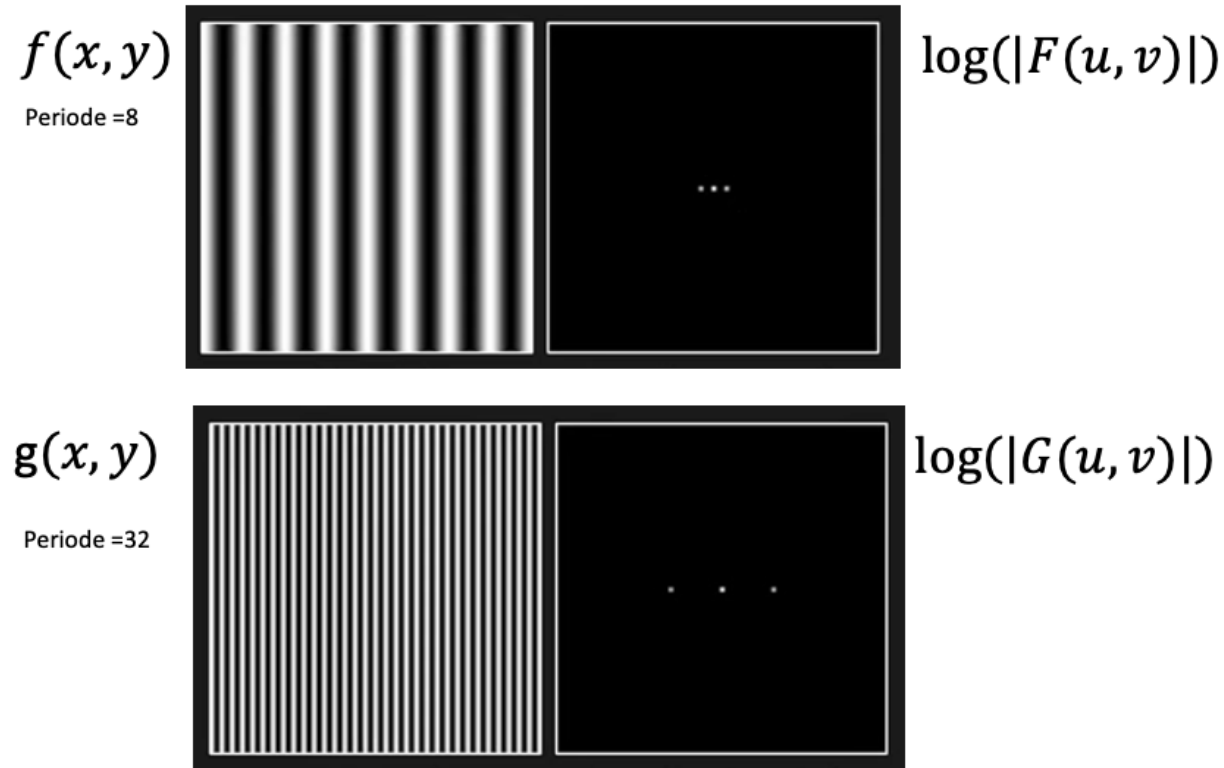
- $f(x, y)$ is a cos function in the x-direction
- We look at the logarithm of the amplitude spectrum:



Frequency domain representation of an image

Examples:

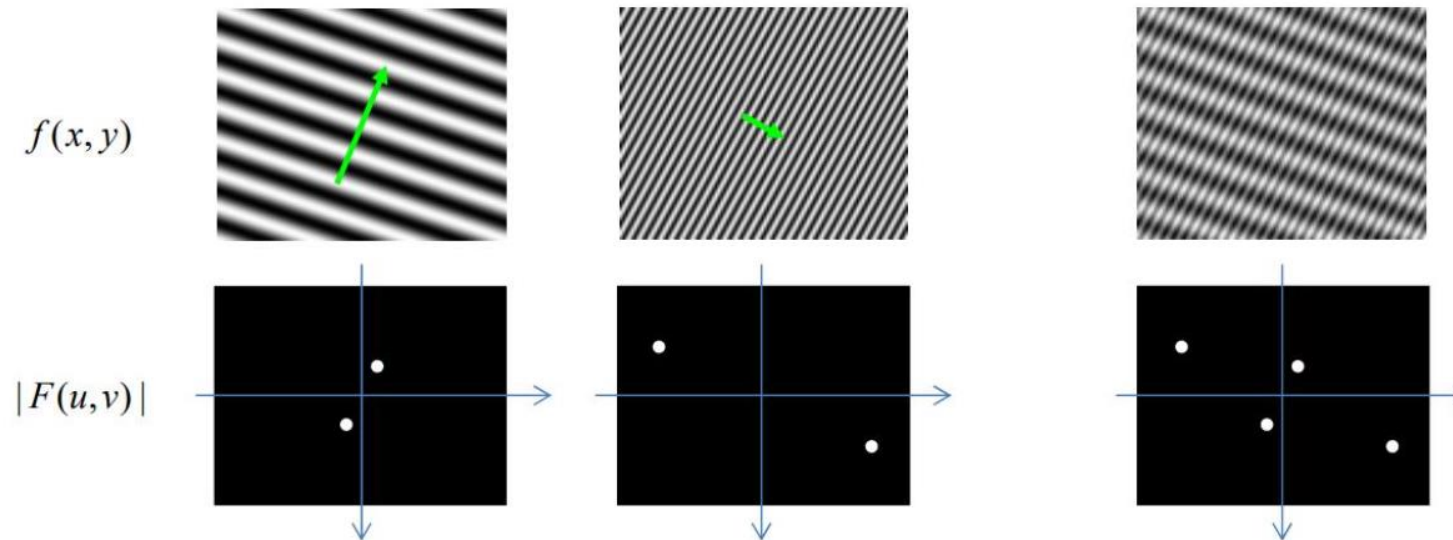
- $f(x, y)$ and $g(x, y)$ are cos functions in the x-direction
- We look at the logarithm of the amplitude spectrum:



Frequency domain representation of an image

Interpretation of the Fourier spectrum

A wave in the image $f(x,y)$ is always linked to a point pair in the amplitude spectrum $|F(u,v)|$. Thus, the distribution of the significant frequencies in the image can be recognized in the amplitude spectrum.



Interpretation of the Fourier spectrum

In the Frequency domain representation, an image is interpreted as sum of harmonic functions with frequency u and v . From the amplitude spectrum conclusions on the frequencies present in the image can be drawn.

